Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/31/2005
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Project Title:	Gravitational Impacts on the Cell Cycle		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	32901	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
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No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 2-1508		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A new cell culture system has been developed specifically for the evaluation of environmental impacts on diverse aspects of the cell cycle of lympho-hematopoietic cells. In this culture system, large quantities of synchronous, newborn cells are generated automatically from cultures of immobilized cells undergoing long-term steady-state growth. The newborn cells are shed continuously from the culture vessels, which can be held stationary or rotated about the horizontal or vertical axes, and used for the cell cycle analyses. The first aim of the study is to assay the effects of gravity averaging on cell cycle phase durations, segregation of components between daughter cells, gene expression, and division plane orientations of mouse and human cell lines cultured in the vessels. The second aim will address the impacts of altered gravity on cell cycle aspects of the differentiation of the human myelomonocytic cell line U937. The third aim will employ the culture system to determine the basic properties of the cell cycle of activated normal human lymphocytes, and changes in the cycle during senescence and during growth in altered gravity. The key to all the studies is that the culture system permits cell cycle analyses on cells undergoing minimally disturbed steady-state growth so any		

	alterations detected in cell cycle properties will be due to altered gravity and not the procedures used to produce the synchronous cells or perform the cell cycle analyses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	An improved version of the rotating bioreactor for use in basic cell cycle research and for analyzing effects of gravity averaging on the cycle during steady-state growth was designed during the fiscal year. As with previous versions, the culture system produces large quantities of synchronous, newborn cells automatically from cultures of immobilized cells undergoing steady-state growth. The newborn cells are shed continuously from the culture vessel while rotating about the horizontal axis and can be used for diverse cell cycle analyses. The culture system in the rotating bioreactor consisted of approximately 50 to 100 million mouse L1210 cells attached to a concanavalin A-coated 100-sq.cm. nitrocellulose membrane filter held vertically in the vessel. Bioreactor experiments were initiated by introducing the cultures by syringe through the exit port of the bioreactor. This approach is made possible by the placement of a flexible support screen behind the coated membrane to enable flushing the cells into the membrane surface without distorting the flat membrane or introducing bubbles that could not be removed. This technique enabled rapid, aseptic initiation of each bioreactor experiment. At this time the direction of culture medium flow was reversed so that the medium flowed past the attached cells and out the exit port such that, at cell division, one new daughter cell was shed from the surface and exited the vessel while the other daughter remained attached to the surface. The rotating bioreactor yielded large quantities (over 4 million cells/hr) of essentially pure, synchronous newborn cells from a culture in undisturbed, steady-state growth. Steady-state growth was achieved because the number of newborn cells released into the effluent of the bioreactor each generation did not vary for multiple generations. This indicates that one daughter cell was typically shed from the immobilized culture at each division. If that were not the case, and occasionally both daughters remained attached, the total number of newborn cells released into the effluent would have increased in successive generations because the number of cells on the surface would have increased. Thus, the cells on the surface, and the released newborn cells, satisfied the definition of steady-state because the condition of the culture did not change over time. Neither the number of cells in the culture vessel nor the composition of the fresh medium bathing the cells changed with time, yielding a stable microenvironment for individual cells. Thus, the key innovation in the current technology is the long-term production of synchronous cells that are in undisturbed steady-state growth. The data from several such experiments demonstrated that rotation had no effect on the fundamental cell cycle properties, such as lengths of cycle phases, as long as the cultures were in steady-state growth. For more detailed cell cycle analyses, the bioreactor has been used in two ways. Firstly, the technique was used for collection of synchronous newborn cell samples and performing standard cell cycle examinations during subsequent synchronous growth. Due to the normal broad distribution of interdivision times in growing cultures, a collection for 10% of the generation time did not produce a measurable spread in the quality of synchrony. Secondly, cell cycle properties and behaviors of the steady-state population in the culture vessel are being examined by performing labeling on the immobilized population and observing the effects in their progeny. As an example of this approach, the cell cycle timing of synthesis of a molecule is determined by pulse-labeling the immobilized population with a precursor and measuring the label on their progeny as they are continuously released. The primary advantage of this approach is that the label is added to undisturbed cells. This method is now being used, in collaboration with Dr. Itamar Simon, Hebrew University, Israel, to determine the sequence of gene replication in the cell cycle and the effects of gravity averaging on this process.
Bibliography Type:	Description: (Last Updated: 07/05/2019)
Articles in Peer-reviewed Journals	Eward KL, Van Ert MN, Thornton M, Helmstetter CE. "Cyclin mRNA stability does not vary during the cell cycle." Cell Cycle. 2004 Aug;3(8):1057-61. PMID: 15254409 , Aug-2004
Articles in Peer-reviewed Journals	Lebleu VS, Thornton M, Gonda SR, Helmstetter CE. "Technology for cell cycle research with unstressed steady-state cultures." Cytotechnology. 2006 Jul;51(3):149-57. Epub 2006 Nov 21. https://doi.org/10.1007/s10616-006-9024-5 ; PubMed PMID: 19002885; PubMed Central PMCID: PMC3449808 , Jul-2006
Patents	US 6,867,040 B2. Awarded March 2005. Mar-2005 Helmstetter CE, Thornton M, Gonda S. "Bioreactor and methods for producing synchronous cells."
Patents	U.S. Patent Application Serial No. 10/080,333. Pending May 2005. May-2005 Helmstetter CE. "Method for producing synchronous cells and the resultant cells."
Presentation	Helmstetter, C.E. "Cell cycle studies with synchronous cultures" Midwest Research Institute May-2004